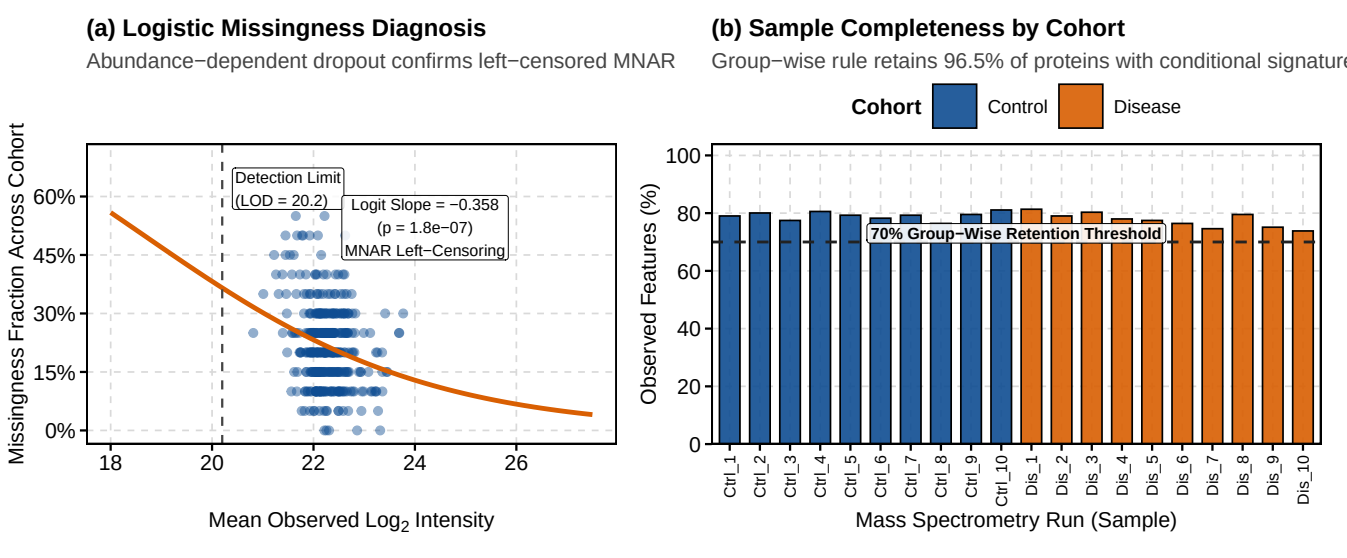
(a) Logistic Missingness Diagnosis
Abundance−dependent dropout confirms left−censored MNAR
Detection Limit
(LOD = 20.2)
Logit Slope = −0.358
(p = 1.8e−07)
MNAR Left−Censoring
Missingness Fraction Across Cohort
60%
45%
30%
15%
0%
18
20
22
24
26
Mean Observed Log2 Intensity
(b) Sample Completeness by Cohort
Group−wise rule retains 96.5% of proteins with conditional signature
Cohort
Control
Disease
70% Group−Wise Retention Threshold
Observed Features (%)
100
80
60
40
20
0
Ctrl_1
Ctrl_2
Ctrl_3
Ctrl_4
Ctrl_5
Ctrl_6
Ctrl_7
Ctrl_8
Ctrl_9
Ctrl_10
Dis_1
Dis_2
Dis_3
Dis_4
Dis_5
Dis_6
Dis_7
Dis_8
Dis_9
Dis_10
Mass Spectrometry Run (Sample)